Corridor or Vehicle?

Comparing Study H with Study H-R — what the relaxed constraints reveal about the faults that had no recovery

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1 The two studies in one paragraph

Study H injected 12 engine plants at 15 points along one Apollo-anchored nominal descent — 180 optimal-control solves — and asked *where on the descent is this fault survivable.* It found 168 landings and **11 injections with no recovery trajectory at all**.

Study H-R took those failures and asked a different question: *is the vehicle lost, or is the corridor too tight?* It re-solved each one at four progressively weaker sets of path constraints, keeping the altitude floor, the actuator bounds and the landing gate fixed throughout.

This document sets the two against each other.

2 The headline comparison

Measure	Study H (baseline corridor)	Study H-R (relaxed)	Change
Injections with a trajectory	169 of 180	171 of 180	+3
Injections that land in the gate	168 of 180	170 of 180	+2
Cases with no trajectory at all	11	9	−2

3 of the 12 carried-forward cases found a trajectory once the corridor was widened, and 2 of those still touch down inside the **unchanged** Apollo gate. The gate was never relaxed — only what the vehicle was permitted

to do on the way there.

3 What the difference actually is

3.1 Reading 1 — the failures were mostly about the corridor

3 cases that Study H reported as unrecoverable are recoverable by a vehicle allowed outside the nominal envelope. Those results were never statements about the vehicle's capability; they were statements about the *problem as posed*. Reported as lost vehicles, they would have overstated the severity of η=0.50, engine out by exactly that margin.

The binding constraint names what each one needed:

Baseline limit the recovery had to break	Cases	Which
glide cone	3	engine_out@38, engine_out@51, thrust_loss_50@46

3.2 Reading 2 — what the relaxation could not fix

9 cases resisted the entire ladder:

Fault	Injected at	Altitude	Range to pad
dead time	17 s	419 m	1746 m
dead time	25 s	282 m	1276 m
dead time	30 s	212 m	978 m
dead time	38 s	109 m	499 m
dead time	55 s	8 m	17 m
engine out	42 s	56 m	264 m
engine out	46 s	19 m	87 m
η=0.15	42 s	56 m	264 m
η=0.15	46 s	19 m	87 m

The last level removes every state constraint **except** $z > 0$ — the cone, the speed box, the attitude and rate limits all go, and the only thing still asked of the vehicle is to stay above the surface. A failure there is as close to "the vehicle cannot do it" as this machinery can get: nothing but the ground is in its way. **These are the cases a fault-tolerance budget should carry as real losses** — and there are 9 of them, not 11.

3.3 Reading 3 — the faults sort into three groups

Group	Faults	What it means
Recoverable with a wider corridor	η=0.50	Every failed injection of this fault yields to relaxation. Its Study H boundary is a corridor artefact.
Mixed	engine out	Some injections are corridor-limited, some are not. The boundary is real but sits later than Study H put it.
Genuinely unrecoverable	dead time, η=0.15	No relaxation helps. The vehicle cannot reach the pad from these states with this plant.

4 Figures, side by side

Study H's outcome grid, with the red cells being the ones this study re-examined:

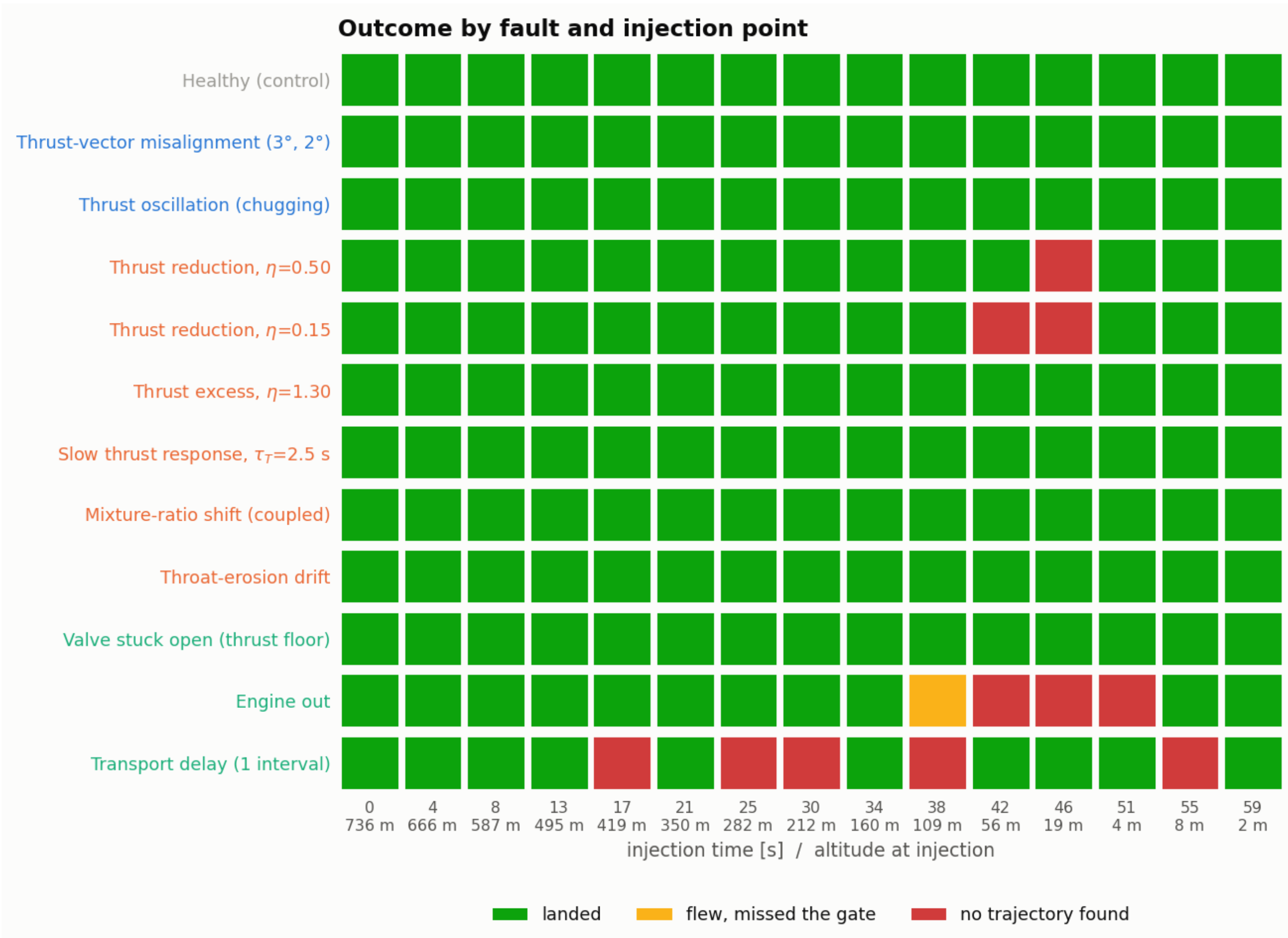

Figure 1: Study H — outcome by fault and injection point

And the ladder result for exactly those cells:

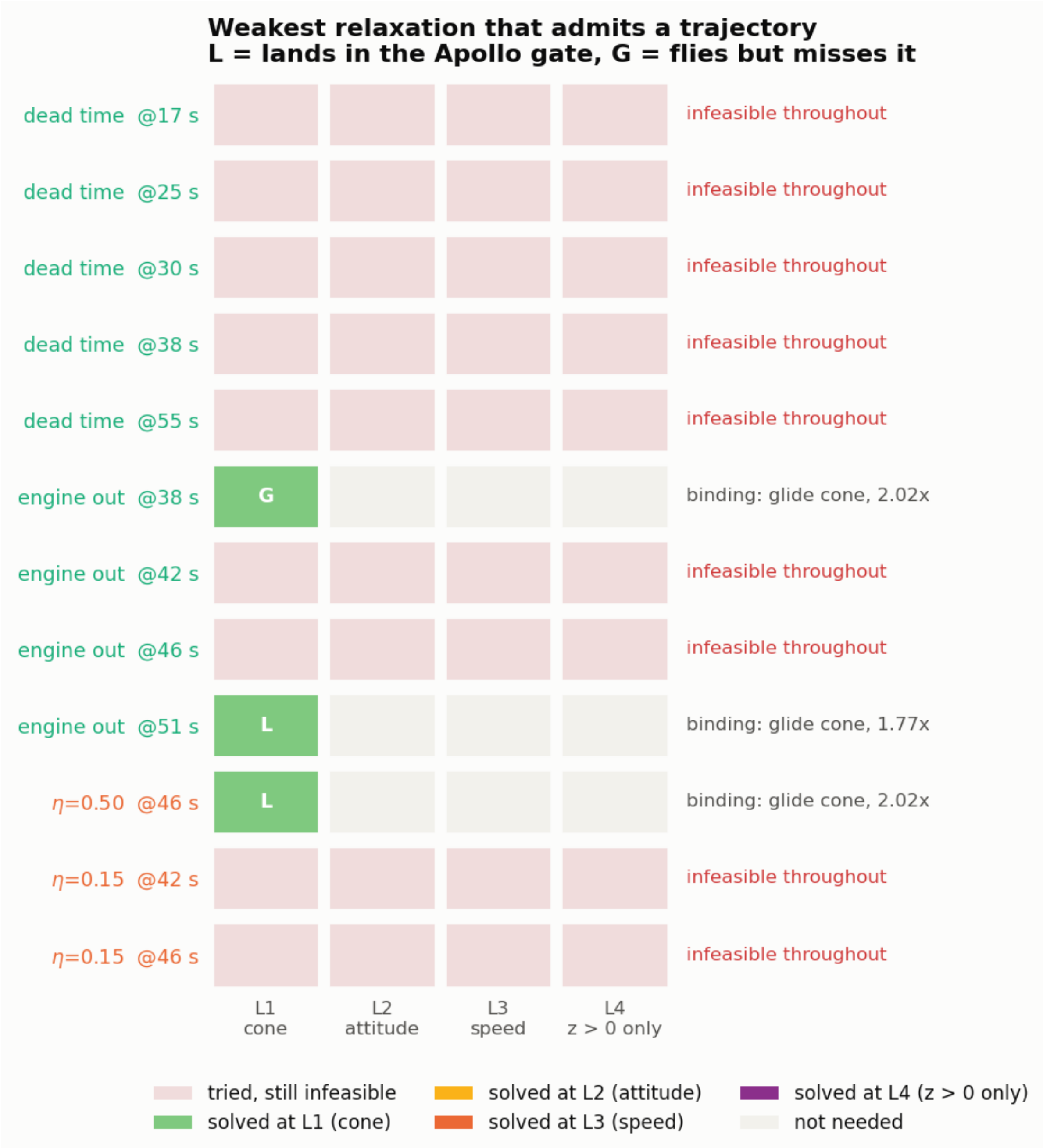

Figure 2: Study H-R — the weakest relaxation that admits a trajectory

The excursion chart is the bridge between them: it shows, for each recovered case, which Study H limit the new trajectory had to break and by how much.

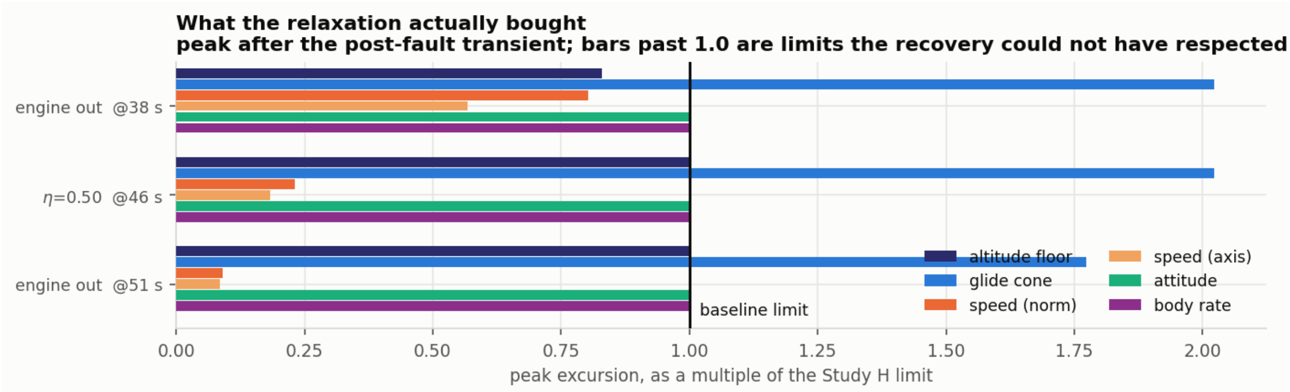

Figure 3: Peak excursion beyond each baseline limit

5 What we can understand from this

A solver's "infeasible" is a property of the constraint set, and it must be reported as one. The single most transferable result here is methodological: 11 failures became 9 once the same states and the same plants were given a wider corridor. Any study that reports fault survivability from a constrained planner is reporting the constraints as much as the vehicle, and the only way to know the split is to vary them deliberately.

Severity should be quoted with the corridor attached. "This fault is unrecoverable after t = X" is incomplete. The defensible statement is "this fault is unrecoverable after t = X *inside a 12° cone with 45° attitude and 60 m/s limits*, and after t = Y with the corridor open" — two numbers, and the gap between them is the value of an envelope-expanding guidance mode.

The floor is never negotiable. Every level of the ladder keeps $z > 0$. Dropping it would let the optimiser route a "recovery" through the ground and still score it against the landing gate, which inspects only the touchdown state — a trajectory that is arithmetic rather than flight. L4 is therefore the weakest honest problem available, and 9 cases fail even it.

Relaxation buys a trajectory, not necessarily a landing. Of the 3 recovered cases, 2 still land in the gate. The rest reach the surface outside it. Feasibility and success are different questions and the distinction survives the relaxation intact — which is what makes the recovered landings credible rather than an artefact of loosened scoring.

The cases that remain are worth more than the ones that moved. After the ladder, 9 cases are left. Those have now survived 22 independent seeds across two studies and four corridors. That is a far stronger claim than the original 11 could support, and it is the number worth defending.

6 Caveats that apply to the comparison itself

- **The two campaigns are not equal-effort by design.** The relaxed run spends more search on fewer cases, deliberately: extra search can only convert a false "no" into a true "yes", never the reverse. It cannot manufacture a landing that does not exist, but it does mean "H-R found more" is partly a statement about effort as well as about constraints.
- **Only the failures were re-run.** The 168 baseline landings were not re-solved under the relaxed corridor, so this document compares the *failures* of one study with the *failures* of another, not two complete campaigns. A relaxed re-run of the landings would be expected to change margins, not outcomes.
- **The relaxed limits are round numbers**, chosen to bracket the baseline rather than derived from vehicle structure. The ladder localises which constraint binds; it does not find the exact threshold.
- **Everything Study H assumed still holds**: instantaneous perfect detection, open loop, one nominal trajectory, constant mass.

7 Reproducing

```
python studies/fault_injection/run_injection_study.py     # Study H campaign
python studies/fault_injection/harden.py                  # H hardening pass
python studies/fault_injection/analyse_injection.py       # H figures + JSON
python studies/fault_injection/run_relaxed_study.py       # H-R ladder
python studies/fault_injection/analyse_relaxed.py         # H-R figures + JSON
python studies/fault_injection/build_comparison_report.py # this document
```